MACRORAIL · 3D SCAN MODULE

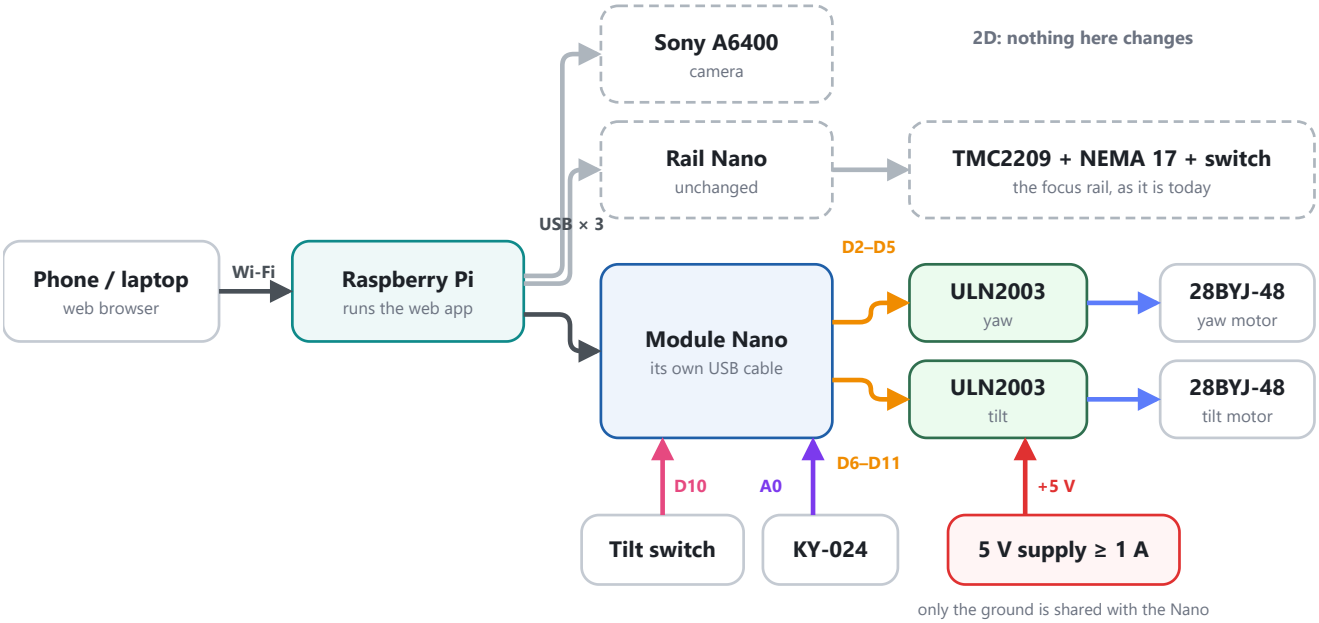
Scan module wiring

Wire the scan module's two motors, its tilt switch and its hall sensor to their own Arduino Nano, and get both axes moving.

1 Overview

The scan module has its **own Arduino Nano**, the **module Nano**, on its own USB cable to the Pi. It drives the **yaw** and **tilt** motors (two 28BYJ-48s, each through a ULN2003 driver board), reads the **tilt switch** and the **KY-024 hall sensor**, and talks to the app over USB. The motors get their power from a **separate 5 V supply**.

The rail stays exactly as it is. The rail Nano, its firmware and its wiring don't change at all. Unplug the module and you have today's rig.

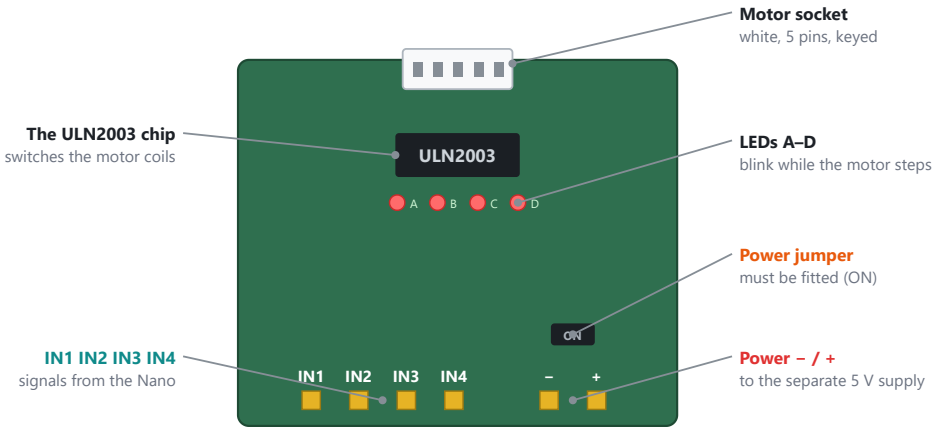


What's in this guide

2	**Meet the parts**	Driver boards, motors, hall sensor, switch, and the shopping list.
3	**Build the circuit**	The module Nano's pins, the complete wiring picture and a numbered checklist.
4	**Cables on the module**	Slack for the tilt, and what to keep clear of.
5	**Flash & configure**	Module firmware, and how the app tells the two Nanos apart.
6	**First power-up checks**	Directions, switch, hall signal, then homing.
7	**Troubleshooting**	What a symptom usually means.

2 Meet the parts

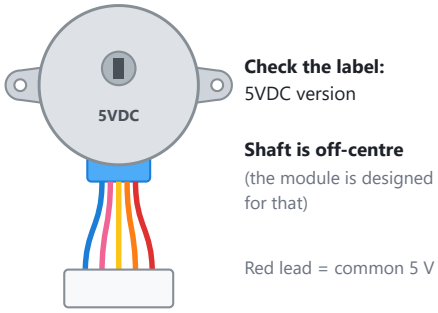
The ULN2003 driver board (×2)



Typical layout. Go by the labels printed next to the pins on your board.

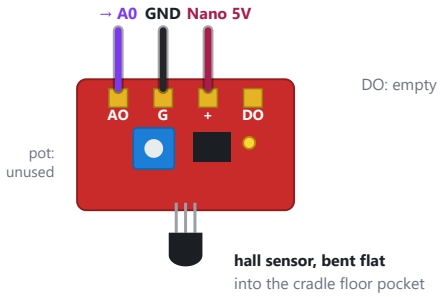
One board per motor. The Nano's pins can't drive a motor coil, so the board's ULN2003 chip switches each coil for it: **IN1–IN4** come from the Nano, the motor plugs into the **white socket**, and **– +** take the separate 5 V supply. The four LEDs show which coils are on. They blink while the motor moves, and they are dark when it stands still, because the firmware switches the coils off 300 ms after every move.

28BYJ-48 motor (×2)

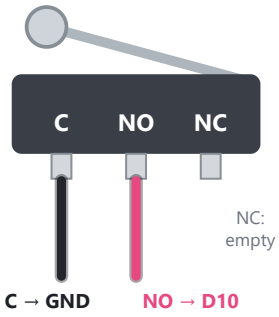


Buy the **5 V** version (*5VDC* on the label); the 12 V one is too weak at 5 V. Its gearbox holds the load with the power off. The plug goes into the board's white socket, and fits only one way.

KY-024 hall sensor



It reads the magnet in the yaw head. Only its **analog output AO** is used: the app finds the peak of the field, which repeats far better than the board's on/off output (DO). So DO, its LED and its potentiometer are unused. It reads about 512 with no magnet near.



Tilt switch (roller microswitch)

Wired like the rail's home switch: **C to GND, NO to D10**, NC empty. Pressing it pulls D10 low. The firmware watches it during every move, so the first contact stops the tilt at once.

Wiring it normally-closed (C + NC) instead? Set `SWITCH_ACTIVE_LEVEL` to `HIGH` in `firmware-module/src/config.h`.

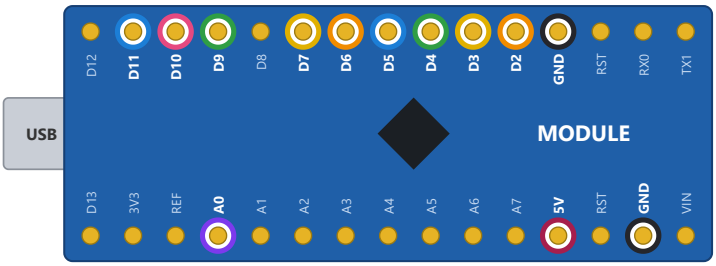
What you need

Part	Notes
Arduino Nano: the module Nano	A CH340 clone is fine. A second one, next to the rail's.
2 × ULN2003 driver board	Usually sold with the motor.
2 × 28BYJ-48, 5 V version	Yaw (under the cradle floor) and tilt (on the −Y upright).
KY-024 linear hall board	Bolted under the cradle floor, sensor bent flat into its pocket.
Roller microswitch	On the printed bracket on the +Y upright.
5 V supply, at least 1 A	For the motors only. A bench supply set to 5.0 V works well.
Small breadboard (optional)	Half size, 30 columns. A terminal block works too: same connections.
Jumper wires (Dupont)	Male-female for the boards, male-male on the breadboard, long enough for the module.
USB cable (Mini-B for most Nanos)	A data cable: a charge-only cable powers the Nano but the Pi can't see it.
Cable ties, multimeter	Ties for the cradle cables; the meter for the checks in section 6.

Why a separate 5 V supply?
Each 28BYJ-48 draws about 240 mA while it moves. The Nano's 5 V comes from the Pi's USB port, which also feeds the camera link and the rail Nano; motor current there causes resets and USB drop-outs. So the motors get their own supply, and only its **ground** is joined to the Nano's. **Never join the supply's + to the Nano's 5V pin.** Two supplies would fight each other.

3 Build the circuit

The module Nano



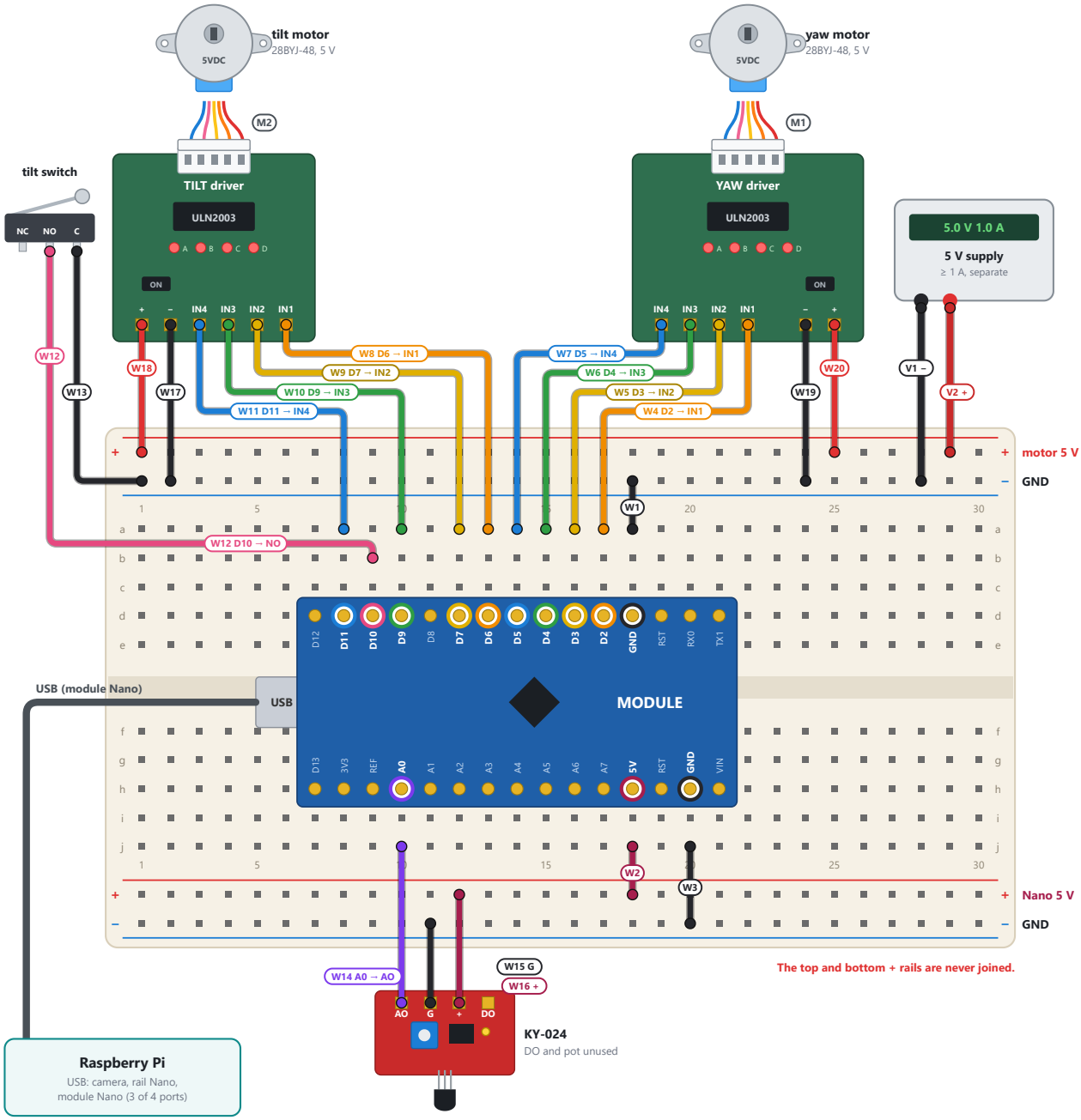
As it sits on the breadboard: USB on the left.

Pins this module uses

- **D2–D5** yaw driver IN1–IN4
- **D6–D11** tilt driver: D6, D7, D9, D11
- **IN1** D2 and D6
- **IN2** D3 and D7
- **IN3** D4 and D9
- **IN4** D5 and D11
- **D10** tilt switch NO
- **A0** KY-024 AO (hall signal)
- **5V** KY-024 + only
- **GND** common ground (both pins)

The yaw driver takes D2–D5 and the tilt driver D6, D7, D9, D11 (D8 stays free, D10 is the switch), both in the order IN1, IN2, IN3, IN4. IN1 has the same colour on both boards, and so on, so a swapped pair is easy to spot.

The complete build



Holes a–e of one column are one strip, f–j another; each rail runs the full length. The **top rails** carry the motor supply, the **bottom rails** the Nano's own 5 V for the hall sensor. The GND rails are joined through the Nano's GND pins; the two + rails never touch. Motors, switch and sensor sit on the module.

Quick reference

Nano	Goes to	Why
D2, D3, D4, D5	Yaw ULN2003 IN1, IN2, IN3, IN4	Yaw motor coils, half-stepped.
D6, D7, D9, D11	Tilt ULN2003 IN1, IN2, IN3, IN4	Tilt motor coils, half-stepped.
D10	Switch NO (switch C → GND)	Low when pressed. Internal pull-up; watched during every move.
A0	KY-024 AO	The magnet's field, 0–1023, about 512 with no magnet near.
5V	KY-024 +	The sensor only. Never the drivers, never the supply.
GND	Both GND rails	Common ground with the motor supply: required.

Wire it in this order

Tick off one line at a time and compare with the picture. “Top rails” are above row a, “bottom rails” below row j. Holes are named row + column, like `a17`. On the driver boards, go by the labels printed next to the pins.

> **Reusing a Nano from another project?** Take off any capacitor between **RST** and GND (often 10 µF, fitted to stop the Nano resetting when its port opens). The app relies on that reset: the Nano then says which controller it is, and uploading firmware needs it too. A capacitor across 5V and GND can stay.

1 · Board and jumpers

☐	**P1**	**Module Nano**	rows **d** and **h**, columns **7–21**	D12/D13 in column 7, USB left
☐	**P2**	**Power jumpers**	on both ULN2003 boards	the ON/OFF jumper by − +

2 · Wires

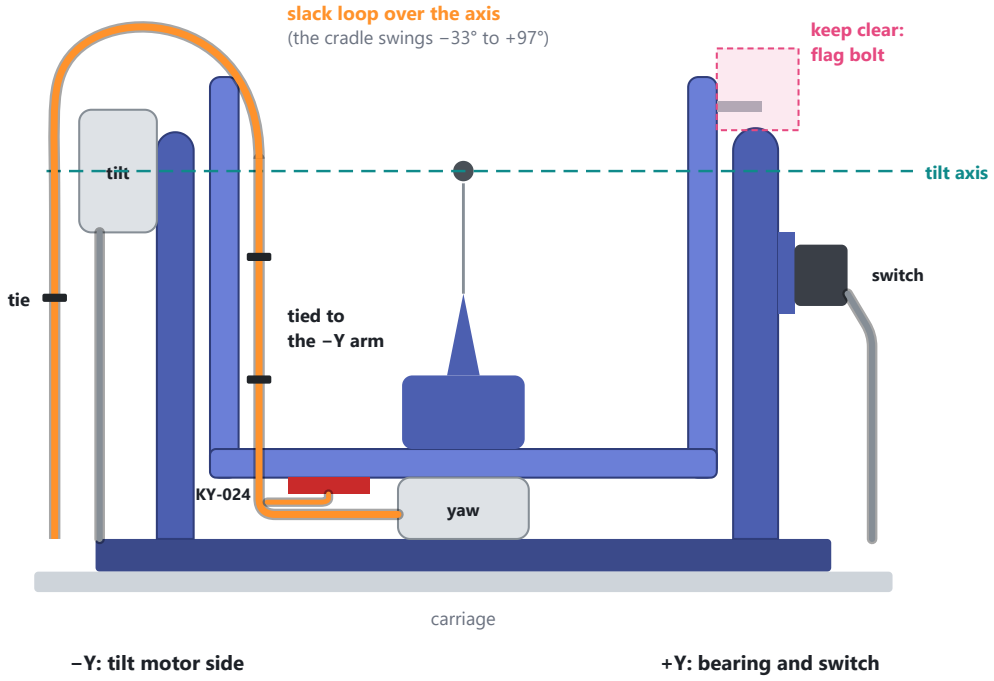
☐	**W1**	a18 (Nano GND, top row)	→	top GND rail	common ground with the motors
☐	**W2**	j18 (Nano 5V)	→	bottom + rail	Nano 5 V, sensor only
☐	**W3**	j20 (Nano GND, bottom row)	→	bottom GND rail	
☐	**W4**	a17 (D2)	→	yaw IN1	
☐	**W5**	a16 (D3)	→	yaw IN2	
☐	**W6**	a15 (D4)	→	yaw IN3	
☐	**W7**	a14 (D5)	→	yaw IN4	
☐	**W8**	a13 (D6)	→	tilt IN1	
☐	**W9**	a12 (D7)	→	tilt IN2	
☐	**W10**	a10 (D9)	→	tilt IN3	D8 stays free
☐	**W11**	a8 (D11)	→	tilt IN4	D10 between is the switch
☐	**W12**	b9 (D10)	→	tilt switch NO	runs between rows a and b
☐	**W13**	top GND rail (column 1)	→	tilt switch C	NC stays empty
☐	**W14**	j10 (A0)	→	KY-024 AO	the analog pin, not DO
☐	**W15**	bottom GND rail	→	KY-024 G	
☐	**W16**	bottom + rail (Nano 5 V)	→	KY-024 +	
☐	**W17**	tilt driver −	→	top GND rail (column 2)	
☐	**W18**	tilt driver +	→	top + rail (column 1)	motor 5 V
☐	**W19**	yaw driver −	→	top GND rail (column 24)	
☐	**W20**	yaw driver +	→	top + rail (column 25)	motor 5 V
☐	**U1**	module Nano USB	→	Raspberry Pi	a data cable, not charge-only

3 · Last: motors and supply, with the supply unplugged

☐	**M1**	yaw motor plug	→	yaw driver socket	keyed: fits one way
☐	**M2**	tilt motor plug	→	tilt driver socket	keyed: fits one way
☐	**V1**	5 V supply −	→	top GND rail (column 28)	supply unplugged
☐	**V2**	5 V supply +	→	top + rail (column 29)	supply unplugged

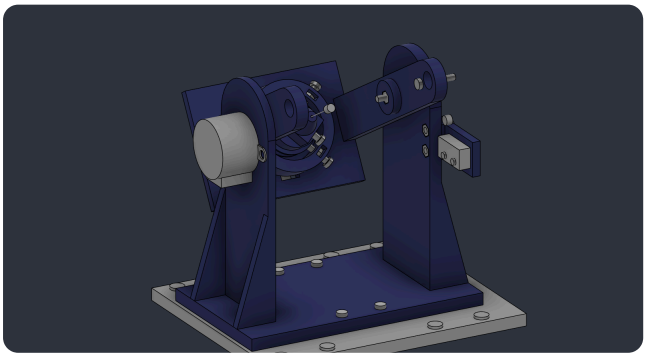
4 Cables on the module

The **tilt motor** and the **switch** sit on the fixed fork, so their cables just run down the uprights. The **yaw motor** and the **KY-024** ride on the cradle, which swings from −33° (belly towards the lens) to +97° (where the switch clicks, past the top view). Their cables must follow that swing without pulling, rubbing or catching anything.

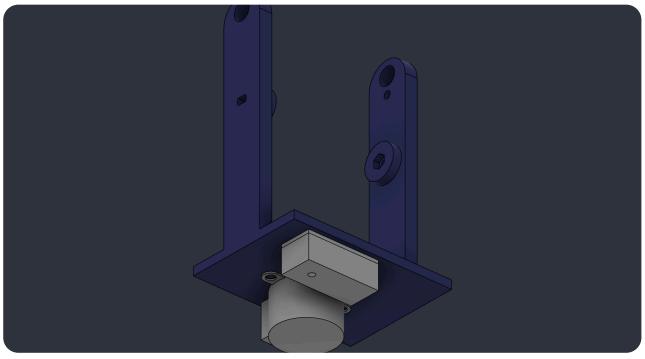


Seen from the camera. Cables to the breadboard leave at the back (−X).

- [ ] **Take the cradle's cables to the −Y side** (the tilt motor side), along the underside of the floor and up the −Y arm. Tie them to the arm with two small cable ties.
- [ ] **Leave a loose loop near the tilt axis,** where the arm moves least. Then run the cables down the *back* (−X) of the −Y upright, next to the tilt motor's cable, and tie them there. The loop must allow the full 130° swing.
- [ ] **Keep the +Y side clear.** The flag bolt on the +Y arm swings past the switch. Nothing may hang there, or it could press the switch or stop the flag.
- [ ] **Keep the camera's side clear.** The lens looks between the uprights from the +X side, so all cables leave the module at the back.
- [ ] **The yaw head turns without limit.** Nothing electrical turns with it, so no cable may touch the head or the centring screws.
- [ ] **Check it before scanning:** after homing (section 6), go to the lowest tilt the app allows, then to +90°, slowly, and watch the loop.



The module seen from the front: tilt motor on the −Y upright (left), switch bracket on the +Y upright (right).



Under the cradle floor: the yaw motor and the KY-024 board. Their leads go to the −Y arm.

5 Flash & configure

Plug in

Plug the camera, the rail Nano and the module Nano into the Pi's USB ports (3 of its 4). The module Nano gets its power from USB; the motors from their own supply.

Flash the module firmware

The app owns the serial ports, so stop it first. On the Pi:

```
sudo systemctl stop macrorail
ls /dev/serial/by-path/      # unplug and replug the module Nano: its entry comes back
~/.local/bin/pio run -d ~/macrorail/firmware-module -e nano -t upload \
    --upload-port /dev/serial/by-path/<the module's entry>
sudo systemctl start macrorail
```

From your PC over SSH: `ssh -t pi sudo systemctl stop macrorail`, flash, then `ssh -t pi sudo systemctl start macrorail`. Any computer with PlatformIO can flash it too: `pio run -d firmware-module -e nano -t upload --upload-port <port>`.

Use `-e nano_new` if the upload fails on a genuine or new-bootloader Nano. Be sure to pick the module's port: flashing the module firmware onto the rail Nano would stop the rail working until you reflash it with `firmware/`.

Tell the app where the Nanos are

Both Nanos are CH340 clones with no serial number, so their `/dev/serial/by-id` names are identical and can't tell them apart. In `~/.config/macrorail/config.toml`:

```
[rail]
port = "auto"

[module]
port = "auto"
```

With `"auto"` the app asks each Nano what it is: the module firmware says so in its first line (`role=module`), the rail's doesn't. If you'd rather fix them, give each its USB socket's path from `ls /dev/serial/by-path/` instead; that path stays the same as long as the cable stays in the same socket. **An old by-id path under `[rail]` must go:** with two Nanos plugged in, it can point at either.

Then restart the app: `sudo systemctl restart macrorail`.

6 First power-up checks

Do these in order, **before any homing**, with no specimen in the holder. Move the rail so the module is well clear of the objective: until tilt is homed, the angle is unknown and no limits apply.

1. **Supply polarity.** Supply off, measure first: set the supply to **5.0 V**; + goes to the top + rail, − to the top GND rail. With a meter on beep, the top GND rail beeps to the Nano's GND; the two + rails **don't** beep to each other or to GND.
2. **Power on.** Supply on, Nano on the Pi. Open the app, switch to **3D**, and check the **Scan module** card on the System tab: it shows *connected*. The Rig tab shows tilt and yaw as not homed. Nothing moves yet, and all driver LEDs are dark.
3. **Tilt direction.** On the Rig tab, jog tilt by **−1°**, a few times. The tilt driver's LEDs blink, and the cradle must tip the insect's **belly towards the lens** (the tilt angle goes down; the flag bolt moves away from the switch). If it turns the other way, set `TILT_INVERT = true` in `firmware-module/src/config.h` and reflash (section 5).
4. **Yaw direction.** Jog yaw by **+10°**. The head must turn **counter-clockwise seen from above**. If it turns clockwise, set `YAW_INVERT = true` in the same file and reflash.
5. **Switch.** Press it by hand. The app shows it *pressed*, and released when you let go.
6. **Hall sensor.** Watch the hall reading on the Scan module card and turn the head slowly past the sensor, by hand (the coils are off when idle) or by jogging yaw 10° at a time. Away from the magnet it reads about 512; near it, the reading must move at least **40** away from that, up or down. If it hardly moves, check that the sensor is bent flat in its pocket and that the magnet is in the head.
7. **Home tilt.** It turns slowly away from the lens until the flag presses the switch at about +97°, backs off until the switch releases, and touches it again slowly. That contact is the tilt reference. Watch the first time, ready to press Stop.
8. **Home yaw.** It turns counter-clockwise for up to about 1.3 turns, finds the magnet's peak, comes round again and stops on it from the same side. At **yaw 0** the magnet is over the hall sensor and centring screw X points at the camera.
9. **Cables.** Go to the lowest tilt the app allows and to +90°, slowly, and check the loop (section 4).

> **Then:** the Rig tab's quick poses (Level, Top view, Screw X to you…) work, and the calibration and centring steps follow in the scan module's assembly and usage guide.

7 Troubleshooting

Symptom	Likely cause and fix
Motor hums or twitches but doesn't turn	IN wires in the wrong order (IN1–IN4 must be D2–D5, or D6, D7, D9, D11, in that order), or a wire loose. Or no 5 V on the driver: check the supply, the driver's − + wires and its power jumper. LEDs that blink while the motor stays still point at the motor plug or the supply.
Nothing moves, LEDs dark	The IN wires don't reach the board, or the module isn't connected (see below).
Turns the wrong way	Set `TILT_INVERT` or `YAW_INVERT` in `firmware-module/src/config.h` and reflash. Don't rewire the motor plug.
Tilt homing fails: NO_SWITCH	The tilt turned all the way without the switch closing. Check the switch wiring (C → GND, NO → D10, not NC) and press it by hand: the app must show it pressed. Check that the flag bolt reaches the roller. If the tilt turned towards the lens instead, the direction is wrong (TILT_INVERT).
Tilt homing fails: SWITCH_STUCK	The switch stayed pressed after backing off. The switch is wired C + NC (reads pressed when open), the flag bolt is set too far out, or the bracket sits too close.
The tilt stops with ENDSTOP during a move	Something pressed the switch: a cable on the +Y side, or a loose switch wire picking up noise. Reroute or resolder.
Yaw homing fails: NO_MAGNET	The field peak was under 40 counts. Check the AO wire goes to A0 (not DO), the sensor's + and G, that the sensor is bent flat under the magnet, and that the magnet is in the head.
Tilt loses steps near +90°	The cradle is heavy there. Homing on the switch shows any loss. Add the counterweight: an 8 mm rod with nuts through the holes at the top of the arms.
The module isn't found	Check `port = "auto"` under `[module]`, that the module firmware is flashed, that no capacitor sits between RST and GND, and that the USB cable carries data (try another). `ls /dev/serial/by-path/` must show it.
The rail stopped being found after plugging the module in	Its config still names a by-id path, which now points at either Nano. Set `port = "auto"` under `[rail]` (or its by-path socket) and restart the app.
The Pi or the Nano resets when a motor starts	The motors are fed from the Nano's 5 V. Move the drivers' + to the separate supply; only the grounds are joined.
LINK errors, motion stops	The module lost contact with the Pi for 2 s: a loose USB cable, or the Pi under-powered. Use the Pi's proper supply.

The full message list is in `docs/protocol-module.md`; the pin constants are in `firmware-module/src/config.h`.